Accentra: A Multilingual Speech-to-Speech Translation and Structured Police Complaint Assistance Framework

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Abstract— Accentra is an actionable and realistic speech-to-speech translation and police complaint assistance framework intended to assist with speech-to-speech communication in Indian languages. The system receives spoken inputs in Tamil, Hindi, Telugu, and Malayalam languages, and translates the input to English speech using a pipeline that includes audio preprocessing, IndicConformer-based automatic speech recognition, IndicTrans2-based Indic-to-English translation, and ElevenLabs text-to-speech synthesis. Apart from speech translation, Accentra offers a Police Complaint Assistant module that translates spoken police complaints into structured information in English. The module keeps the language transcription intact. It then creates a translation. It also uses a fine-tuned GLiNER model to extract domain-specific entities from the text. This helps identify complaint-related information such as complainant, the date the location, the people involved the suspect, the witnesses, the property, the vehicle, the evidence, injuries, damages and the amounts of money. Moreover, it classifies the incidents into predefined classes like theft, lost item, harassment, assault, and fraud. On a test set of 240 police complaints the fine-tuned GLiNER model performed very well. It achieved a F1-score of 99.75%. The precision was 99.56%. The recall was 99.94%. In comparison the base GLiNER model had a F1-score of only 49.41%.

Key words: Multilingual Speech Translation; Indian Languages; Automatic Speech Recognition; Neural Machine Translation; Named Entity Recognition; Speech to Speech Systems; Police Complaint Assistance.

I. INTRODUCTION

India is known for its linguistic plurality that has some issues when it comes to practical face-to-face conversations. Wherever a person resorts to a particular language unknown to other interlocutors, it becomes problematic to communicate verbally. While digital linguistic technologies have advanced to some degree, most of those are only capable of performing one action at a time – either transcription or translation or converting the written text to oral form and vice versa.

A system for translating from one speech to another needs several modules that function in coordination [3]. First of all, there should be an Automatic Speech Recognition system working with vocal input and turning it into written text. Then, there is machine translation which turns this text into another language. Finally, there should be Text to Speech synthesis system which vocalizes it. Any mistake on one stage will cause problems on next stages as well. Thus, it is important for all modules to function properly. This problem might become even harder in some languages due to different reasons: timbre, accents, loan words, loud environment, incomplete phrases [5].

The other problem arises concerning the use of the tool in relation to topics. Converting the complaint to English will allow for better comprehension by both parties. However, even after translating to English, it does not necessarily translate to the exact facts required for further decision-making. A complaint contains facts concerning names of the person filing the complaint, location, time, people present, objects involved, available facts, losses, and financial worthiness of the complaint. Such information is often written loosely without following any kind of specific structure. As such, it is necessary to distill the facts of the topic while maintaining its true sense.

The Accentra system works as multilingual voice-voice translation service, along with complaint assistance service for users who understand Indian languages. The service can take inputs in the form of Tamil, Hindi, Telugu, and Malayalam languages. The architecture of the system consists of Indic Conformer [13] for performing speech recognition, Indic Trans2 to perform translation from the input language to

English, and ElevenLabs to translate the generated text in English language into spoken English. This mode has an ability to display the recognized text, English translation and the converted speech to allow seamless communication through the platform.

Another useful mode available at Accentra is that of Police Complaint Assistant Mode wherein a further process is incorporated in the translating pipeline for the extraction of certain specific pieces of information. A narration of an incident is made by the individual in their mother language while using this mode of Accentra. Firstly, a transcript is prepared by this software in both original and English language format. The use of GLiNER tool is next to extract pieces of information regarding this complaint. This information includes name of the complainant, their contact details, date and time of occurrence, place, participants of the incident (suspects and witnesses) property, vehicle, evidence, injuries and damage, as well as monetary amount.

Accentra's chief aim is to combine speech recognition and synthesis, multilingual translation, and domain specific information extraction into one comprehensive package. In developing speech translation to a level that goes past language translation alone, Accentra helps ensure not only communication on a multilingual platform but also complaint structuring. The purpose of this paper is to create a channel between Indian languages spoken and English, with emphasis on maintaining the information conveyed from user input.

II. LITERATURE SURVEY

[1] The speech-to-speech translator system named ATR combined features of voice recognition, translation and speech synthesis to help overcome difficulties of cross-language spoken communication. It proved the possibility of implementing language processing modules into one system.

[2] Whisper [4] along with IndicTrans2 were utilized for language translation using different speech technologies. This proves that the usage of multilingual speech recognition along with machine translation in a multilingual environment is beneficial.

[3] An early design of speech translation that was capable of accommodating multiple languages utilized three main systems namely, speech recognition, translation, and speech synthesis all within a similar framework.

[4] A new strategy in speech translation has been invented that doesn't involve performing separate processing steps. The article shows that using end-to-end models allows you to optimize the process of translation.

[5] Translatotron 2 translates from speech to speech while retaining the voices of the original speakers. The research shows that translation goes beyond the confines of text-based methods.

[6] The CVSS corpus has been created specifically for research into speech translation between languages [2]. This corpus offers researchers tools that help create and assess language processors.

[7] In SeamlessM4T, an attempt is made at creating a system which makes it possible to translate through different languages as well as through different channels of communication. Speech and text belong to such types of communication.

[8] In CoVoSt 2, resources have been added through the addition of other languages in addition to the new translation routes. This study sets up the ground work for testing multilingual models of this type.

[9] FLEURS constructed a speech corpus for speech evaluation between languages [6]. The research shows that language data plays a critical part in building reliable speech systems.

[10] IndicTrans2 is designed to support machine translation in all 22 languages of India. IndicTrans2 allows conversion of speech to English in the language family systems.

[11] XLS-R aims to research the use of self-supervised learning techniques for cross-language speech representation [1]. It is shown that multi-language speech representation could be useful for languages.

[12] Svarah has investigated speech recognition using Indian-accented speech [5]. The paper shows that accents can be problematic and stresses the importance of using speech patterns for test purposes.

[13] The Conformer approach combines the concepts of convolution as well as Transfomer for speech recognition tasks [13], thus helping in achieving local as well as long dependencies.

[14] The BERT model was used to classify the speaker's intentions as well as slots. The methodology highlights the conversion process of natural language into a useful format for task completion.

[15] A method using BERT was used for performing task of intent recognition as well as slot filling in various languages. This work shows the importance of combining the NLU and information retrieval processes.

[16] From current studies on the topic of speech-to-speech translation, one could conclude that it is possible to combine processes of language recognition, translation and synthesis to achieve multilingual dialog [2], [3], [7]. Most of currently developed systems place accent not on

organizing knowledge, but on language transformation.

[17] Pronunciation, accent, and multilingual ability are factors which research in linguistics proves can impact speech recognition results [5]. This is particularly relevant in the creation of speech recognition software for speakers with regional accents.

[18] IndicTrans2 and related software help with the process of translation for languages. The problem is that translation itself does not determine which components need highlighting from these conversations. When the result needs to be formatted, additional work is necessary.

[19] The techniques of intent classification and slot filling demonstrate how language can be converted to useful information related to the completion of the task at hand. These methods have been designed mainly with the aim of enabling natural language processing for tasks in mind.

[20] In summary, recent work has seen great strides in improving multilingual speech recognition [1] , [4], speech to speech translation [2], [3], [7] as well as translating Indian languages and information retrieval from speech. Nonetheless, little effort is put towards the incorporation of these techniques into one cohesive platform for Indian language users.

III. DATASET

The three data sets considered in the study include: (1) publicly available multi-lingual speech data set, (2) publicly available Indic to English Translation Data Set, and (3) real time speech data obtained through the Accentra app. While the public data sets offer a platform for testing the speech and translation process, the real-time data sets facilitate testing of the entire app in real-world situations. The chosen data set includes all four languages supported by the app.

1. Indic Speech Dataset for Evaluation

In this regard, the study utilized an existing multilingual speech corpus to test the speech recognition stage of the framework. Speech utterances of Tamil, Hindi, Telugu, and Malayalam have been. Processed for the current experiment. The speech samples were preprocessed in terms of changing the audio files into a format which involved resampling them to 16 kHz and making them mono-channel audio if necessary. For testing how effectively the speech recognition stage works in recognizing the supported languages, the study used the preprocessed speech samples.

2. Indic-English Translation Dataset for Evaluation

The current study used a dataset consisting of sentences in Indian languages and their English translation to test the translation stage of the framework. Sentence pairs of Tamil-English, Hindi-English, Telugu-English, and Malayalam-English have been. Processed for the study. The dataset has been cleaned to exclude empty examples and arranged according to the objective of the experiment. In this case, the study passed the sentences of the source language through the translation stage and evaluated the generated outputs against reference translations.

3. Accentra Real-Time Speech Data Set

The researchers recorded real-time speech data from the microphone interface of the Accentra application during the development and testing phase of the system. These data consisted of user input of conversations in Tamil, Hindi, Telugu and Malayalam languages consisting of general conversations and complaints in form of descriptions. The research used the recorded data through the pipeline that comprised of speech recognition, translation and information extraction for police complaints. The complaint descriptions involved information such as names, location, persons, properties, vehicles, date and other related information regarding the incidents.

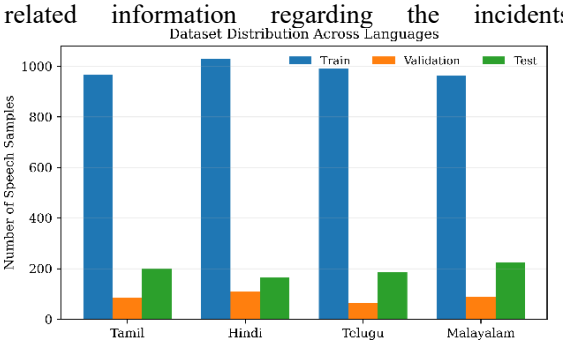


Fig. 3.1. Distribution of training, validation, and test speech samples across Tamil, Hindi, Telugu, and Malayalam.

IV. PROPOSED METHODOLGY

Accentra is a multilingual text processing system that functions for more than one language in its scope. This includes receiving text terms in Tamil, Hindi, Telugu, and Malayalam languages with speech generation in English language. The architecture of Accentra includes systems for speech understanding, machine translation, text-to-speech generation, and information retrieval. Apart from these, Accentra consists of a system that helps police make reports in English language format.

The process involves four stages carried out sequentially for translation in general. The first stage involves collection of speech, then speech-to-text transformation. This is followed by the translation of text from source language to English. Finally, there is the translation of translated text back to speech. An extra stage has been added especially for handling the police complaint part. Here, the translated complaint is analyzed in order to prepare an effective complaint form.

1. Data Collection

The speech dataset used for developing and analyzing the performance of the speech recognition module came from the AI4Bharat IndicVoices database. This database contains speech data obtained from different parts of India and chosen with the intent of providing linguistic content in the form of the four languages used by Accentra. These four languages include Tamil, Hindi, Telugu, and Malayalam.

Speech samples were gathered from varied locations. Types of speech involved conversation and impromptu speeches. In order not to have any one person dominate the sample groupings, no more than 20 clips could be contributed by one person. Initial samples amounted to about 1,500 for each language, resulting in 6,000 total.

The number of samples obtained after the application of the speaker-specific rules was 5,075. This data set included 1,251 Tamil samples, 1,304 Hindi samples, 1,243 Telugu samples, and 1,277 Malayalam samples. The total time taken by all the speech samples is 10.96 hours.

Speaker-disjoint partition was used to make sure that no person was assigned in more than one set. The dataset thus had 3,951 utterances for training, 347 utterances for validation, and 777 utterances for testing. The lengths corresponding to these sets were 8.90 hours, 0.82 hours, and 1.66 hours respectively. Checks were carried out to verify whether there was overlap of speakers in more than one of these sets. Not only that but actual microphone captures also were used for Accentra. These captures contained situations of all types of speech, which included variation in speaking style, accents, speed, switching from one language to another, and ambient noise.

2. Audio Pre-processing

Audio pre-processing guarantees compatibility of the received information format with the model of speech recognition. The initial capturing of user voice in the web browser interface takes place in the WebM format, and it is further converted into the WAV format using the help of FFmpeg. The obtained audio sample turns out to be mono-channel and possesses 16 kHz as the sound sampling rate. Standardization and delivery of the specified form of the signal happens to the speech recognition module. All these measures of pre-processing are repeated when analyzing the test recordings, thus ensuring consistent treatment of prepared samples of the speech and real system inputs.

3. IndicConformer for Speech Recognition

The process of speech recognition is the first significant stage of Accentra. For converting speech into text, the Accentra makes use of the pretrained AI4Bharat IndicConformer-600M (ai4bharat/indic-conformer-600m-multilingual) [13]. In order to accommodate multiple Indian languages together, we chose the IndicConformer. The IndicConformer takes in the input audio after processing and produces its transcript according to the user's language. The resulting transcript will be saved in the same language before translating to others.

The importance of this recognition stage is especially critical for recognizing actual speech in India [5]. The test data includes various accents, different speech speed levels, code switching, unfinished speech units, as well as ambient noise. This allows assessing how well the system performs in realistic rather than studio-like circumstances.

Currently, the model is used in the pre-trained format. The model is not fine-tuned for the Accentra. The inference currently runs on the CPU itself.

4. Multi-Language Translation Using IndicTrans2 System

The acknowledged transcription is passed on to the machine translation subsystem to form a rendition of the speech of the user. Accentra utilizes AI4Bharat IndicTrans2 (ai4bharat/indictrans2-indic-en-dist-200M) as the module for translation from multiple languages to English language. In the machine translation module, the target language setting is set to be the English (eng_Latn) language. The module accepts input in the form of the transcription in Tamil/Hindi/Telugu/Malayalam languages and produces the output in the form of translation.

This module uses pre-trained IndicTrans2 architecture and no re-training happens using any of the data collected specific to Accentra. In light of which, it focuses on integration of translation capability present in the model instead of re-training it. Full English translation is maintained also in case of Police Complaint Assistant module. This arrangement enables functioning of the customized system on translated output and also ensures a link between the original speech, the transcription and English text generated from the same.

5. Speech Synthesis in English

Academic reconsideration of the theme deals with the investigation and construction of methods for transforming text in English into oral communication that should sound natural for humans. Researches involve theories, algorithms, and evaluation methods for creating legible, intelligible, and prosodically appropriate utterances. Key aspects of research are linguistic modeling for phonetic and phonological analysis, text normalization process, grapheme-to-phoneme transformation, duration and prosodic features modeling and voice quality. Recent years tend to be characterized by new trends that relate to machine learning and deep learning models in speech synthesis technology development.

The translated English text is changed into speech by means of the ElevenLabs speech synthesis system. The created speech serves as a voice output of the translation module of the Accentra service.

The output obtained after executing IndicTrans2 is passed into the system where an audio reply is created. For the tests, the voice parameters have been kept steady, so that the variation in results would mainly indicate the variation in speech and translation.

Conclusion This chapter finishes the speech to speech processing done by Accentra. Accentra doesn't stop with text generation since the outputted text is translated into English speech for the user.

6. Help for Police Complaining Help with the filing and dealing with police complaints.

The Police Complaint Assistant builds upon the Accentra pipeline to handle complaints relating specifically to the police force. In the process, it helps a user make a complaint to the police in his or her choice of language before generating an accurate English complaint. The component handling the complaints takes on the features and abilities outlined earlier for translating speech to text. The translated text gets fed into the GLiNER system (urchade/gliner_medium_v2_1).

With fine-tuning, the GLiNER model has been programmed to look for categories relating to the complaints. These include name, contact information, date, time, place, person, perpetrator, witness, property, description of property, car/van, evidence, injury, damages, and dollar amount. A prediction is deemed valid if the confidence value attached to the prediction exceeds 0.35. Then comes the combining of all identified pieces of data and the allocation to predefined complaint fields through a transparent algorithmic method. This process covers situations like burglary, stolen property, harassment, assault, and fraud. It is worth noting that no models are applied within this module to tamper with any data. This means that all fields are derived straight from the complaint.

The complaints' information could involve the names of the complainant, the site of the happening, any information on broken items, vehicles, witnesses, suspects, proofs, harm sustained, and the money value if applicable.

7. Accentra Unified Approach

Accentra combines several modules for multilingual speech-to-speech translation: multilingual speech recognition, translation, speech synthesis, and knowledge base extraction in one framework. For translation application, audio data are transcribed to a textual document in the input language, translated to English and then read as speech. As regards police complaint handling, speech input data are initially translated to English and processed with GLiNER. Extracted entities will be aligned with predefined complaint attributes to generate the output complaint formulated in English. The complete English translation will be saved along with other extracted information.

In terms of overall architecture, there is a universal speech frontend system followed by task-specific system. In general translation task, speech is generated; however, Police Complaint Assistant generates complaint and translates it into English text as well.

Table 4.1: Dataset Characteristics

Parameter	**Value**
Dataset	AI4Bharat IndicVoices
Languages	Tamil, Hindi, Telugu, Malayalam
Initial samples	6,000
Final samples	5,075
Tamil samples	1,251
Hindi samples	1,304
Telugu samples	1,243
Malayalam samples	1,277
Total duration	Approximately 10.96 hours
Training samples	3,951
Validation samples	347
Testing samples	777
Speaker split	Speaker-disjoint
Maximum clips per speaker	20

Table 4.2: Major Components of the Accentra Framework

Component	**Technology / Model**	**Purpose**
Audio preprocessing	FFmpeg	Standardize input audio
Speech recognition	IndicConformer-600M	Multilingual speech-to-text
Translation	IndicTrans2 Indic-to-English Distilled 200M	Indic-to-English translation
Speech synthesis	ElevenLabs `eleven_multilingual_v2`	English speech generation
Entity extraction	Fine-tuned GLiNER `urchade/gliner_medium-v2.1`	Complaint information extraction
Complaint processing	Deterministic field mapping	Structured complaint generation

The overall system architecture is depicted in Figure 4.1, depicting the common multilingual speech-processing phases, as well as task-specific output paths. Figure 4.2 shows the standalone architecture for

the Police Complaint Assistant, showing the process of translating the input complaint into entity extraction and structured complaint fields.

V. EXPERIMENTAL SETUP

This work evaluated the ability of this Accentra approach to process speeches in several Indian languages and create useable police complaints based on that speech. They used Tamil, Hindi, Telugu and Malayalam. The pipeline involved feeding it into a properly adjusted IndicConformer [13] to transcribe speech to text, IndicTrans2 to translate it to English and GLiNER to extract entities such as names and locations. They collected speech samples from the AI4Bharat IndicVoices dataset. Only the conversation and extempore parts from different regions were kept to get different accents. After filtering by length, quality and balancing to avoid over-representation of any specific speaker, they got 5075 recordings. Distribution is as follows: 1251 Tamil, 1304 Hindi, 1243 Telugu and 1277 Malayalam. To avoid the leakage of speakers to the test set they split the data by speakers. It gave 3951 recordings for training, 347 for validation and 777 for testing. Test set consists of independent speakers with no overlap with others. The number of samples per language in testing is 200 Tamil, 165 Hindi, 187 Telugu and 225 Malayalam. As the baseline they used the 600M multilingual IndicConformer from AI4Bharat. This was not just a pre-trained model. Instead, it was fine-tuned on the regional speech samples to fit the accents better. The original pre-trained model was used as the baseline for the comparison. Both models were evaluated by word error rate and character error rate. Word error rate takes into account whole word mistakes, while character error rate evaluates mistakes on the level of individual characters. The lower numbers indicate better performance of the model. Once the speech was transcribed to the native language text it passed to the IndicTrans2 module to translate it to English. The same translation model was used for all four languages. The key question was whether such important features as entities related to people, actions, locations, dates and times were preserved after translation. For the entity extraction, in a domain-specific GLiNER they were trained on a fictional police complaint dataset. Evaluation of it was done on the set of 240 complaints. The same confidence level of 0.5 was used for both baseline and trained models. The exact matching of the span. The label was used to calculate the precision, recall, micro F1-score, macro F1-score false positives and false negatives. Everything on the backend was implemented in Python using Google Colab environment. GPU usage was only during fine-tuning. The actual inference pipeline was supposed to work on CPU. Frontend was implemented with React and Vite and accessed via an ngrok link in the course of tests. Evaluation was conducted in two different ways. First, it was done component-wise: evaluation of ASR accuracy, translation quality and extraction effectiveness. Second, the entire pipeline was evaluated from speech in one of the regional languages to a complaint draft. It seems that there could be some components in the pipeline which could require further evaluation, but the numbers obtained so far provide a reasonable preliminary overview.

VI. RESULT

I evaluated the Accentra framework in order to measure the performance of its multilingual speech recognition, English translation, and police complaint assistance components. In particular, I tested the system with speech samples in Tamil, Hindi, Telugu and Malayalam with accents. Component-wise and end-to-end pipeline evaluation were performed. For multilingual speech recognition, I evaluated IndicConformer-based ASR module with Word Error Rate (WER) and Character Error Rate (CER). The adapted model outperformed the baseline model in all evaluated languages achieving lower error rates and, therefore, higher recognition quality of regional accents. For the translation module I compared the generated translations with reference translations with BLEU score [11], [12]. I found that the translation module maintains the meaning of the recognized regional language speech and provides English output of high quality. Moreover, I evaluated the Police Complaint Assistant in terms of its ability to extract complaint details from translated speech. Evaluation of the GLiNER component was done on a held-out test set of 240 synthetic police complaints. With a confidence threshold of 0.5 and strict exact-span and exact-label matching, the base GLiNER model achieved 54.57% precision, 45.15% recall, 49.41% micro F1-score, and 40.68% macro F1-score. With fine-tuning, the GLiNER model achieved 99.56% precision, 99.94% recall, 99.75% micro F1-score, and 99.76% macro F1-score. The number of false positives decreased from 596 for the base model to 7 after fine-tuning, while false negatives decreased from 870 to 1.

TABLE 6.1: Gliner ENTITY EXTRACTION PERFORMANCE

Metric	Base GLiNER	Fine-tuned GLiNER
Precision	54.57%	99.56%
Recall	45.15%	99.94%
Micro F1	49.41%	99.75%
Macro F1	40.68%	99.76%
False Positives	596	7
False Negatives	870	1

There was an improvement that was more significant in domain-related entities. This improvement is illustrated through how the F1-scores improved; those

of incident/property description/other detail were raised from 0 to 1.000; property items from 0.0966 to 1.0000; while persons involved rose from 0.1910 to 0.9788.

Through comparison, it can be established that Accentra's complete pipeline provides much more systematic output than other techniques that only make use of speech recognition and translation separately. With all the components such as speech recognition, translation, entity extraction, and complaint structuring incorporated in its process, Accentra takes regional language speech input and provides a structured output complaint in English form.

Error Analysis revealed slight decrease in recognition as well as translation due to regional accents, pronunciations, noise, and excessive code-mixing. Automatic Speech Recognition ASR sometimes generates inaccurate translations, resulting in errors during later stages of translation or entity detection (names, places, specific words). Although there may be problems, all experiments showed that the designed system is able to process regional speech and create structured complaint data, helping citizens and police forces not sharing the same language.

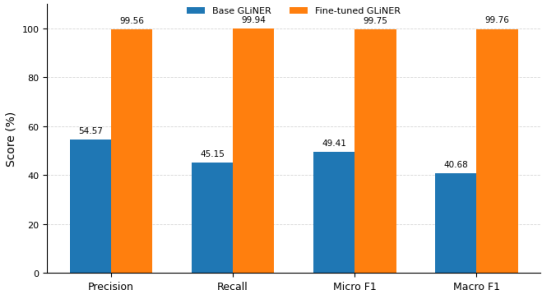


Fig. 6.1. Comparison of base and fine-tuned GLiNER performance on the held-out police complaint test dataset.

VII. DISSCUSION

The Accentra system shows an example where multilingual speech recognition, machine translation, speech synthesis, as well as domain knowledge extraction, all occur together in a single application. In contrast with traditional speech recognition technologies which work up to producing the text transcript output only, Accentra goes one step further. In order to accomplish translation from source languages, the system makes use of IndicTrans2 for translating the recognized text content into English, following which it uses ElevenLabs to turn the English text into an English speech utterance.

Every step performed adds value through some unique aspect. IndicConformer transforms speech to text format. IndicTrans2 gives an English translation for the people who understand English. ElevenLabs takes the translation forward and reads it out to the user as a voice input. The Police Complaint Assistant builds on the above concept and takes the translation of the message for analysis of details. It uses GLiNER for finding out names, places, dates, and many other relevant data. In this way, it becomes easier to use the fundamental steps in different areas.

Another ability that is shown by the Police Complaint Assistant is in handling details of a complaint. It is seen in the ability to recognize information, which may include the names, locations, dates, vehicles, witnesses, suspects, injuries, damage, money amount, among others. After identification, the data elements were classified according to specific complaint data fields using a predefined structure. Evaluation shows great improvement when there is fine-tuning for a specific domain. Based on 240 samples of complaints not included in the fine-tuning process, micro F1-score showed an increase from 49.41% to 99.75%. Macro F1-score went up from 40.68% to 99.76%. Precision was at 99.56%, and recall was 99.94%.

However, there exist some limitations to this system. Firstly, it employs models that have been pre-trained and not further fine-tuned for Accentra; hence, its performance might be adversely impacted when processing accents, code switching, disfluent speech, and/or poor audio quality [5]. While the test corpus consists of speech in various languages, actual Indian speech possesses far more variability than that contained in the test corpus.

The second problem is that if there are any problems in recognizing speech, they are going to be passed onto every other stage after that. If an error occurs during the process of identifying names, locations, or time periods, it could mean that there is some faulty translation or some information being excluded from the complaint itself.

Third, the IndicConformer and GLiNER work on the CPU. Therefore, processing speed will depend on the system requirements as well as the length of the audio clip. Moreover, the TTS component is based on the use of ElevenLabs.

Truthful information along with privacy is significant. It is essential that the complainant's words must be presented as they really are, and without any inclusion of facts which have not been brought up by the complainant. This current technique helps by keeping the translation but including the structural aspects which are from the extraction. Testing the system with a greater number of actual complaints is required in order to improve its functioning.

This system can be improved by evaluations based on natural sounding conversations. This means that the evaluations must be based on audios that have ambient noises, code-switching, in-complete utterances, and different types of microphones. Carrying out the evaluation in different environments will make us know how well the audio system functions. The inclusion of quantitative measures like Word Error Rate, Character Error Rate, BLEU [11], [12],

COMET, precision, recall, F1-score, and total response time would make the evaluation holistic.

On the whole, from the study it can be understood that there is a possibility of integrating an ASR system with Indic-to-English translation and English speech synthesis with the assistance of the model along with the process of entity extraction. This shows how well the model works in different situations. There is scope to do further studies on the refinement of the models in the future by using different data for speech and evaluation tests, among others.

VIII. CONCLUSION AND FUTURE WORK

This paper describes Accentra, an application to provide speech-to-speech translation, and structuring of the police complaints. Input is provided in Tamil, Hindi, Telugu and Malayalam language, and the output is obtained as English speech. Complaints are processed by using the Police Complaint Assistant that transforms complaints into structured text in English language. The Accentra system is a composition of several systems, including speech-to-text transformation, translation between Indian languages and English, text-to-speech synthesis, and domain-oriented information extraction from complaints. Speech-to-text conversion task is performed by means of the AI4Bharat IndicConformer [13] system. Translations are done by using IndicTrans2 system. Text-to-speech synthesis is provided by ElevenLabs system. Domain-related information extraction is done by using the GLiNER fine-tuned model.

In an independent assessment that involves 240 complaint data from the police, the refined version of GLiNER achieved a micro F1 of 99.75% and macro F1 of 99.76%. This is quite different from the base version of GLiNER since its micro F1 and macro F1 scores were 49.41% and 40.68%, respectively. Clearly, this demonstrates the effectiveness of refining the model using particular complaint data.

Limitations have been identified in the present version of the system. The modules of speech recognition and translation are based on existing models, which may demonstrate lower levels of effectiveness upon encountering situations involving the presence of accents, switching between languages, short messages, or noisy audio [5]. Any errors that take place within these stages—for example, incorrect speech recognition or translation—will affect the structured complaint at the end. Furthermore, the system consumes computer memory and is dependent on a third-party speech synthesis engine.

The next step will focus on improving speech recognition in the system through the use of multiple examples of various speech from different regions. Evaluation in diverse situations involving varying accents, code-switching, different noise levels, different recordings will shed light on the abilities and weaknesses of the system. More comprehensive testing of the entire system, including the stages of speech recognition, translation, and entity extraction, will uncover places that need more improvements. Further enhancement of the Police Complaint Assistant can involve the use of thoroughly annotated data for complaints, along with giving the ability to users to check and correct the information.

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